PHY · Electromagnetism — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Coulomb's law

Syllabus: S26-U4-SU-01 / S27-U4-SU-01

Frequency: 4 of 10 exam years; 21 marks in all.

Marks reward: Select the principle or relationship 6, Substitution and working 4, Qualitative explanation 4, Diagram 3, Final answer 2, Recall 1, Units, significant figures, direction 1

Past questions: 2016R-9; 2017P-20(a); 2017P-20(c); 2020R-6; 2024P-13(bi); 2024P-13(bii); 2024P-13(ci); 2024P-13(cii)

Summary: Coulomb's law links the electrostatic force between two point charges to the product of their charges and the inverse square of their separation, and examiners use it both as a direct calculation and as one force within an equilibrium or tension problem. It usually appears in a short calculation, a force-direction statement, or a multi-part question where charged objects hang from strings or push apart and students must explain how the electric force changes the tensions.

Recognise: Look for two charged objects (point charges, balls, or charged spheres) whose charges are given or must be worked out, a question asking for the force, field strength, or equilibrium, and a distance that is either stated or must be found from string length and angle. Questions that ask whether objects attract or repel, or why a tension changes, also point to this point.

Steps

1. Identify the two charges and their magnitudes, converting any microcoulomb or nanocoulomb values to coulombs before using them.
2. Find the separation r between the centres of the charges in metres; if it is not given directly, use trigonometry from the string length and angle first.
3. Substitute into $F = q_1 q_2 / (4\pi\varepsilon_0 r^2)$, using the permittivity of free space from the formula sheet and squaring the distance, not the raw r.
4. Evaluate the magnitude and state it with its unit (newtons), keeping sensible significant figures.
5. Decide the direction: like charges repel and unlike charges attract, and the force acts along the line joining the two objects.
6. For an equilibrium or tension problem, draw a labelled free body diagram, resolve forces horizontally and vertically, treat the electric force as one component, and solve for the unknown tension or angle.

Where the marks are

- A mark for stating Coulomb's law in correct form, with the constant or permittivity, before substituting.
- A mark for a correct substitution with the distance squared and in metres.
- A mark for the final magnitude with its unit, and a mark for the direction (attraction or repulsion) tied to the object it acts on.
- Marks for a labelled free body diagram and resolving forces when the electric force sits inside an equilibrium problem.
- Marks for explanations that link the type of charge to attraction or repulsion and then to a change in tension or position.

Pitfalls

- Using r instead of r squared, or forgetting to square a distance that has been calculated with trigonometry.

- Leaving the distance in centimetres or millimetres rather than converting to metres.
- Giving only a magnitude with no direction, or stating a direction without naming whether it is attraction or repulsion.
- Taking the string length as the separation of the charges instead of the distance between the charge centres.
- Ignoring charge redistribution when charged objects touch: opposite charges partially neutralise and then share the remainder, so the new charges must be calculated first.

Pattern 2: Electric fields around charges

Syllabus: S26-U4-SU-02 / S27-U4-SU-02

Frequency: 2 of 10 exam years; 4 marks in all.

Marks reward: Recall 2, Qualitative explanation 2

Past questions: 2016R-1；2021C-21(b)；2021C-21(c)

Summary: This point covers the electric field that point charges and charged objects set up in the space around them, and the idea that an electrostatic force on a charge is explained by the field at that charge's location. It is examined mainly as short recall of field-diagram features and as explain-style answers linking charge separation to field strength and force.

Recognise: Look for questions asking about field diagrams or field lines, what a charge or charged body does to the space around it, or why a force acts on a charge without contact. Also look for explain questions where charges are separated (for example in a strip or a pair of plates) and you must say what field that separation creates and how it changes with the amount of separation.

Steps

1. Identify the source of the field: a point charge or charged body, and whether charge has been separated so that one region is more positive or negative than another.
2. State that the source alters the surrounding space so that a field exists at every point nearby, whether or not another charge is present to feel it.
3. Use the field definition E = F/q, where F is the force on a small positive test charge q placed at that point, so the field direction is the direction of that force.
4. For a force on a charge in a field, use F = qE, and say the force is caused by the field at the charge's position, not by the source directly touching it.
5. When charge separation is involved, link more separation (larger charge difference or closer spacing) to a stronger field, and no separation to no field.
6. On a field diagram, read the direction from the arrow on the lines (away from positive, toward negative) and the strength from how closely the lines are spaced.

Where the marks are

- A mark for naming a property shown on a field diagram, such as the direction of the field or how strong it is (shown by line spacing or density).
- A mark for stating that the field is what exists in the space around the charge and what carries the electrostatic force to another charge.
- A mark for a causal link: charge separation produces the field, and a stronger separation gives a stronger field.
- A mark for a clear chain of reasoning when the question says explain, for example: no separation, therefore no field, therefore no force.

Pitfalls

- Describing the force as acting at a distance with no mention of the field that carries it.
- Saying the field only exists when a test charge is placed in it, rather than existing around the source regardless.
- Drawing or describing field lines as the paths that charges travel along.

- Stating that a larger separation or magnitude is needed without saying it increases the field strength, or mixing up strength with direction.
- Giving a one-word answer to an explain question instead of linking the cause (charge separation) to the effect (field and force).

Pattern 3: Force on a charge in an electric field

Syllabus: S26-U4-SU-03 / S27-U4-SU-03

Frequency: 6 of 10 exam years; 25 marks in all.

Marks reward: Select the principle or relationship 9, Substitution and working 7, Final answer 4, Diagram 2, Qualitative explanation 2, Recall 1

Past questions: 2018R-5; 2019C-19(e); 2019R-7(a); 2019R-7(b); 2020P-13(b); 2020P-13(d); 2020R-6; 2021C-21(c); 2021C-21(d); 2021C-21(e); 2021P-14(b); 2021P-14(ci); 2021P-14(ciii); 2023P-18(ai); 2025P-18(b); 2025P-18(c)

Summary: This point tests the definition of electric field strength as force per unit charge, and the electric force a charged body feels in a field, including its direction for positive and negative charges. It usually appears as a short calculation, a free-body or force diagram, or a problem where the electric force balances or combines with weight, tension or a magnetic force (parallel plates, oil drops, pendulums, velocity selectors).

Recognise: Any question involving a charged particle, droplet or sphere in a field, a potential difference across a gap between parallel plates, or a field strength value to convert to a force. Key phrases include force per unit charge, field strength, equilibrium, and a charge placed between two plates.

Steps

1. Identify the charge q, including its sign, and convert it to coulombs. Note whether the body is a point charge, a droplet or a particle such as an electron or proton.
2. Write the definition $E = F/q$, so $F = qE$. For uniform fields between parallel plates, also use $E = V/d$, giving $F = qV/d$.
3. Decide the direction: the force on a positive charge is along the field, and on a negative charge it is opposite to the field.
4. If other forces act (weight, tension, magnetic force), draw a free-body diagram and balance forces along each axis, for example tension plus electric force equals weight when the electric force is upward.
5. Substitute values in consistent SI units and calculate. Keep a sensible number of significant figures and include the unit.
6. For a balance involving a magnetic field, equate the electric and magnetic forces ($qE = qvB$) and rearrange to find the required quantity.
7. Check that the sign, direction and magnitude of the answer are physically reasonable.

Where the marks are

- A mark for stating the relationship $E = F/q$ or $F = qE$ (or $E = V/d$) before substituting numbers.
- A mark for a correct substitution with the charge in coulombs and consistent units, shown as an explicit calculation.
- A mark for a final answer with the correct magnitude and unit, stated clearly.
- Diagram marks for drawing the electric force in the correct direction relative to the field, with other forces such as weight labelled.
- Explanation marks for linking the field to its effect on the force and motion, for example a velocity component changing because a force acts along the field.

Pitfalls

- Forgetting that the force on a negative charge points opposite to the field, and so reversing the direction.

- Leaving prefixes such as micro, milli or kilo unconverted, so the answer is off by a power of ten.
- Omitting the weight or another force from the free-body diagram, or not showing the electric force opposite to the weight when it is.
- Mixing up the charge in coulombs with a count of electrons or using the wrong sign in the balance equation.
- Giving a final answer without a unit or with an unjustified number of significant figures.

Pattern 4: Work, potential energy and potential difference

Syllabus: S26-U4-SU-04 / S27-U4-SU-04

Frequency: 9 of 10 exam years; 29 marks in all.

Marks reward: Select the principle or relationship 11, Final answer 9, Substitution and working 7, Qualitative explanation 2

Past questions: 2016P-14(e); 2017C-21(ai); 2017C-21(ci); 2018C-19(a); 2019C-19(d); 2020C-20(di); 2021R-11(a); 2021R-11(b); 2021R-11(c); 2022C-18(d); 2022R-9; 2024P-12(b); 2024P-15(c); 2024P-15(eii); 2025C-20(c); 2025P-15(a)

Summary: This point covers the work an electric field does on a charge as it moves between two points, and the link V = W/q (or ΔV = W/q) that turns that work into a potential difference. It usually appears as a charged particle such as an electron, proton or ion being accelerated (or stopped) by a known voltage, with the answer asked as a kinetic energy, a speed or a voltage, sometimes given in electron-volts.

Recognise: Look for a charged particle (electron, proton, ion) that is accelerated or slowed by a voltage between plates, grids or dees. Key words include accelerating potential, stopping voltage, kinetic energy gained, and eV. Any question asking for the speed after crossing a potential difference, or the voltage needed to reach a given speed, belongs to this point.

Steps

1. Identify the charge q (for an electron or proton use e = 1.60×10^{-19} C, multiplied by the ion's charge number), the mass m, and the potential difference V.
2. Write the energy statement first: the gain in kinetic energy equals the work done by the field, ΔEk = W = qV (equivalently, the potential energy lost).
3. If the energy is given in electron-volts, convert it: 1 eV = 1.60×10^{-19} J.
4. Write kinetic energy as ½mv² and rearrange for the unknown: V = mv²/(2q) or v = √(2qV/m).
5. Substitute in SI units (V, kg, C, m s^{-1}) and calculate.
6. For a stopping voltage, the initial kinetic energy equals the charge times the stopping potential: Ek = eV_stop.
7. Give the answer with units and a sensible number of significant figures. Use the Newtonian formulas unless the question asks for relativistic effects.

Where the marks are

- A mark for stating the energy equation (kinetic energy gained = work done = qV) before any substitution.
- A mark for a correct rearrangement of ½mv² = qV for the unknown quantity.
- A mark for using the correct charge and mass for the particle, including the ion's charge number.
- A mark for converting eV to joules correctly when the energy is given in electron-volts.
- A mark for the final value with units. Explain parts need the electrode charges or field direction linked to the force on the charge.

Pitfalls

- Dropping the ½ in kinetic energy, or squaring the velocity incorrectly when rearranging.
- Using the electron mass for a proton or ion, or ignoring the ion's charge number.
- Mixing up kV and V, or forgetting to convert eV to joules.
- Treating the potential difference as an energy: forgetting to multiply by q, or dividing when the step

should multiply.
- Using relativistic formulas when the question says to ignore relativity, or the reverse.

Pattern 5: Uniform field between parallel plates

Syllabus: S26-U4-SU-05 / S27-U4-SU-05

Frequency: 5 of 10 exam years; 15 marks in all.

Marks reward: Substitution and working 5, Final answer 5, Select the principle or relationship 3, Qualitative explanation 2

Past questions: 2018R-5; 2020P-13(c); 2021C-21(b); 2021C-21(c); 2021C-21(d); 2021C-21(e); 2021P-14(a); 2023P-18(ai); 2025P-18(b)

Summary: This point covers the uniform electric field between two charged parallel plates, using E = ΔV/d to find the field, the force on a charge in it, or sometimes the resulting acceleration. It usually appears as a short calculation within a larger multi-part question on charged particles, with occasional brief explanations of why the field is uniform.

Recognise: Look for two parallel plates (or a strip or gap) with a stated voltage across them and a separation d, plus a charged particle, droplet or ion placed between them. Key words: field strength, potential difference, plate separation, force on the particle, acceleration. Also watch for questions that give the field and ask for the voltage, or ask you to derive a formula from a given text.

Steps

1. Write down the potential difference ΔV across the plates and the separation d, the perpendicular distance between them.
2. Convert everything to SI units first: volts, metres. Change mm or cm to m and kV to V before substituting.
3. Use E = ΔV/d. The field is uniform, so the same value applies everywhere between the plates. Its direction runs from the positive plate towards the negative plate, perpendicular to both plates.
4. If a force is needed, use F = qE = qΔV/d, with q as the magnitude of the particle's charge. Give the direction of the force separately, taking the sign of the charge into account.
5. If an acceleration is asked for, combine with Newton's second law: a = F/m = qE/m, using the particle mass.
6. State the answer with its unit (V m^{-1}, equivalently N C^{-1} for field; N for force; m s^{-2} for acceleration) and a sensible number of significant figures.

Where the marks are

- A mark for writing the relationship E = ΔV/d (or F/q = ΔV/d) before substituting values.
- A mark for substituting correct SI values, with distance and voltage converted to base units.
- A mark for the final numerical answer with the correct unit, and the direction where the question asks for one.
- For an explanation, a mark for a clear cause-and-effect link, such as a uniform field giving a constant force on a charge and a voltage that rises with field strength over the same separation.
- For a derivation, a mark for identifying that the plate separation d is the same quantity as the strip width, then rearranging the equations to the required form.

Pitfalls

- Leaving the separation in mm or cm, or the voltage in kV, which gives an answer off by a factor of 10 or 1000.
- Using a length that is not the perpendicular plate separation, such as a plate length or a total strip thickness.
- Forgetting that the force on a negative charge points opposite to the field, or dropping the minus sign from the charge when reasoning about direction.
- Stating the field unit inconsistently, for example writing V without the per-metre part, or giving N

without the per-coulomb form when the question asks for N C^{-1}.

- Using E = ΔV/d for a point or non-parallel arrangement, where the field is not uniform, or writing the answer without the direction when the question asks for it.

Pattern 6: Conventional current and electron flow

Syllabus: S26-U4-SU-06 / S27-U4-SU-06

Frequency: 1 of 10 exam years; 1 marks in all.

Marks reward: Diagram 1

Past questions: 2019R-1(ai)

Summary: This point asks you to show the direction of conventional current, which is the direction positive charge would move, and to recognise that electrons move the opposite way. In past papers it has appeared as a single diagram question on a circuit, where you add a current arrow to a labelled diagram (only one such part is in the evidence so far, so treat the pattern as thin).

Recognise: Look for a circuit or device diagram with a battery, cell, or power supply and a request to draw or label the direction of current. Words such as 'conventional current', 'direction of current flow', or 'show the flow of positive charge' signal this point. If the question asks about electrons, the answer is the reverse of the conventional arrow.

Steps

1. Identify the terminal of the source that is positive and the one that is negative, using the symbol or polarity marking on the battery or cell.
2. Remember that conventional current leaves the positive terminal of the source and travels through the external circuit back to the negative terminal.
3. Draw a single arrow on a wire of the external circuit, pointing along that path from positive towards negative (outside the source, current runs from + to -).
4. If the question asks for electron flow instead, reverse the arrow: electrons go from the negative terminal through the external circuit towards the positive terminal.
5. Check the arrow sits on the external circuit wire, is clearly labelled if a label is requested, and points the correct way.

Where the marks are

- One mark is awarded for an arrow drawn correctly on the circuit showing conventional current direction, which is the direction from positive to negative through the external circuit.
- The mark depends on the arrow's direction, not its position, so a clearly drawn arrow pointing the correct way on any external wire is enough.
- Examiners look for the arrow to point from the positive side towards the negative side, so a reversed arrow loses the mark even if drawn neatly.

Pitfalls

- Drawing the arrow pointing towards the positive terminal, which is the direction of electron flow, not conventional current.
- Confusing the two conventions and drawing the electron arrow when conventional current was asked for, or the reverse.
- Drawing the arrow inside the battery or on the wrong component rather than on the external circuit wire.
- Misreading the polarity of the source symbol, especially when the long and short plates are easy to confuse.

Pattern 7: Magnetic fields of currents, solenoids and electromagnets

Syllabus: S26-U4-SU-07 / S27-U4-SU-07

Frequency: 2 of 10 exam years; 10 marks in all.

Marks reward: Qualitative explanation 8, Recall 2

Past questions: 2017C-22(a); 2017C-22(c); 2017C-22(d); 2022P-16(d); 2022R-7(b)

Summary: Current-carrying conductors and moving charges produce magnetic fields, and this is the principle behind solenoids and electromagnets. Past questions on this point are mostly short explain or infer items, often set in an unfamiliar context such as planetary fields, a moving conducting bar, or the transmission of AC power.

Recognise: Look for wording about a current or moving charged particles creating a magnetic field, or a device (solenoid, electromagnet, transformer, a bar that moves when a circuit is switched on) where the field is the link to the answer. Context-based questions often hide the current inside a natural process, such as moving molten metal or ionised gas, so identify the moving charge first.

Steps

1. Identify the moving charge: a current in a wire, or moving ions or electrons in a fluid or plasma.
2. State the principle: a moving charge or current produces a magnetic field around it.
3. Link the field to the situation: the field lines circle the wire, and a changing current gives a changing field.
4. For a solenoid or electromagnet, state that a coil of current gives a field like a bar magnet, strengthened by an iron core.
5. For an effect such as a moving bar or a distorted field, say the field interacts with another field or magnet to produce a force or a change.

Where the marks are

- One mark is usually given for the core principle stated correctly, such as a moving charge producing a magnetic field.
- A further mark is given for applying the principle to the context, for example linking moving molten iron to a current and so to a field.
- Extra marks go to describing how the field arises or changes, such as an alternating current giving an alternating magnetic field.
- Ferromagnetic materials are credited when the question asks why a metal core or conducting material matters.

Pitfalls

- Stating that the field comes from the wire alone without saying the moving charge causes it.
- Saying the field is 'produced by electricity' without mentioning moving charges or current.
- Forgetting to connect the field to the specific context, so only the general principle is credited.
- Confusing the field of a single wire with the stronger field of a coil, or omitting the iron core in an electromagnet.
- Giving a one-word answer when the question asks to explain, which usually earns only part of the mark.

Pattern 8: Magnetic field diagrams

Syllabus: S26-U4-SU-08 / S27-U4-SU-08

Frequency: 9 of 10 exam years; 29 marks in all.

Marks reward: Diagram 13, Select the principle or relationship 4, Recall 3, Qualitative explanation 3, Final answer 3, Units, significant figures, direction 2, Substitution and working 1

Past questions: 2016R-1; 2017R-3(a); 2017R-4(a); 2018R-10(aii); 2018R-10(b); 2019C-20(ci); 2019C-20(cii); 2020R-4(b); 2021P-19(b); 2021P-19(c); 2022R-7(b); 2023P-13(di); 2023R-7(c); 2025R-1(a)

Summary: Magnetic field diagram questions ask students to draw or interpret the direction and relative strength of the magnetic field around permanent magnets, straight current-carrying wires and the Earth. They usually appear as sketch or arrow-drawing items worth a good share of the marks, sometimes linked to a short explanation or a calculation using the Earth's field components.

Recognise: Look for wording such as 'draw', 'sketch', 'show the field', 'indicate the direction of the field at point P', or a diagram of a magnet or wire with a current direction given. Questions about the Earth's field with a dip angle, or questions asking which of several diagrams is correct, also belong here.

Steps

1. Identify the source. A permanent magnet has field lines leaving the north pole, curving around the outside and entering the south pole. A straight wire has circular field lines centred on the wire. The Earth's field lines emerge near the geographic south and enter near the geographic north.
2. For a wire, apply the right-hand grip rule: the thumb points along conventional current and the curled fingers show the direction of the circulating field. The strength falls with distance, B = mu0 I / (2 pi r), so the lines are closer together nearer the wire.
3. For a magnet, draw the lines densest at the poles, spreading out and becoming sparser further away, forming the familiar dipole pattern.
4. Draw closed, continuous lines with arrowheads. Lines never cross or touch, and their spacing shows relative strength.
5. For a composite field, add the field vectors from each source at several points, then draw the resultant lines through those points. Check that the shape is symmetric where the setup is symmetric.
6. For an Earth-field angle problem, resolve the total field B into a horizontal component BH and a vertical component BV, using BH = B cos(theta) and BV = B sin(theta), where theta is the dip angle. Label these on a vector diagram before doing the algebra.

Where the marks are

- Correct arrowhead direction, consistent with the current direction or the magnetic pole polarity. This is often a separate mark from the line shape.
- Line density or spacing drawn so that it shows relative strength, for example denser near a wire or a pole.
- Overall shape matches the source: concentric circles around a wire, a dipole-like pattern for a magnet, and an Earth-like pattern with lines leaving near one geographic pole and entering near the other.
- Explanation questions: a clear link between non-uniform field and changing distance from the source, or a statement of which way a charge or object would be pushed.
- Earth-field component questions: a correct trigonometric relationship stated and applied, with a labelled vector diagram showing the components.

Pitfalls

- Drawing arrowheads the wrong way, especially for a wire where the right-hand grip rule is misapplied or the view (front or top) is confused.
- Drawing field lines that cross, touch, break off or spiral. Field lines must be closed, continuous and non-intersecting.
- Drawing evenly spaced lines when the field is clearly stronger close to the source.
- Sketching only the field of one source in a composite question instead of adding the vectors at each point.
- Mixing up sine and cosine when resolving the dip-angle components, so the horizontal and vertical parts are swapped.

Pattern 9: Magnetic flux density of a current

Syllabus: S26-U4-SU-09 / S27-U4-SU-09

Frequency: 6 of 10 exam years; 26 marks in all.

Marks reward: Substitution and working 8, Final answer 7, Select the principle or relationship 6, Qualitative explanation 4, Graph work 1

Past questions: 2017C-22(b); 2017R-13(a); 2018R-10(ai); 2018R-10(aii); 2020R-4(a); 2021P-19(a); 2023P-13(a); 2023P-13(c); 2023P-13(dii); 2025R-1(b)

Summary: This point covers the magnetic flux density around a long straight current-carrying wire, $B = \mu_0 I / (2\pi r)$, where the field strength falls as the perpendicular distance from the wire increases. It usually appears as a short calculation of field strength for a given current and distance, sometimes with a diagram and a combined-field vector question, and occasionally as an explanation of why the field is not uniform.

Recognise: A long straight wire or conductor carrying a steady current, a perpendicular distance from the wire, a compass or test point placed nearby, or a graph of B against 1/r where the gradient must be used to find the constant μ.

Steps

1. Identify the current I (in amperes) and the perpendicular distance r from the wire to the point of interest.
2. Convert r to metres before substituting, since centimetres and millimetres are common traps.
3. Use $B = \mu_0 I / (2\pi r)$ with $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}, or the value given on the data sheet.
4. Calculate B, give it in tesla, and round to the significant figures the question requires.
5. If a graph of B against 1/r is given, its gradient equals $\mu_0 I / (2\pi)$, so rearrange to find $\mu_0 = 2\pi \times$ gradient / I.
6. For a resultant field, find each field's magnitude and direction separately, then add them as vectors: perpendicular components combine by Pythagoras and the angle comes from $\tan\theta$.
7. Use the right-hand rule for direction: thumb along the conventional current, fingers curling in the direction of the field.

Where the marks are

- Writing the equation or a correct rearrangement and substituting values earns the principle mark.
- The unit conversion of distance is usually a separate calculation mark, so show it explicitly.
- A correct final value with units and sensible significant figures earns the answer mark.
- Combined-field questions award marks for each individual field, for the resultant magnitude, and for the direction stated as an angle with a reference direction; a labelled diagram can also earn marks.
- Explanation questions reward stating that B is inversely proportional to r, that B changes with position, and therefore that a non-uniform field gives a path that is not circular.

Pitfalls

- Substituting centimetres or millimetres directly without converting to metres.
- Dropping the 2π in the denominator, or using μ_0 where $\mu_0/(2\pi)$ is needed.
- Using the distance along the wire instead of the perpendicular distance r.
- Giving only the magnitude of a resultant field and omitting its direction.
- Stopping at the gradient on a B against 1/r graph without rearranging to find μ_0.

Pattern 10: Magnetic force on moving charges and wires

Syllabus: S26-U4-SU-10 / S27-U4-SU-10

Frequency: 10 of 10 exam years; 81 marks in all.

Marks reward: Qualitative explanation 28, Final answer 16, Substitution and working 13, Select the principle or relationship 12, Diagram 8, Units, significant figures, direction 4

Past questions: 2016C-21(di); 2016C-21(dii); 2016P-17(c); 2016R-10; 2016R-5; 2017C-21(d); 2017C-22(e); 2017R-3(b); 2017R-4(b); 2018C-19(d); 2018C-19(e); 2019C-19(e); 2019C-19(f); 2019C-20(b); 2019P-15(eii); 2019R-1(aii); 2019R-1(b); 2020C-20(a); 2020C-20(c); 2020C-20(dii); 2021C-21(b); 2021C-21(c); 2021C-21(e); 2021R-3…

Summary: This point covers the force on a moving charge or a current-carrying conductor in a magnetic field: its direction, its size from $F = qvB\sin\theta$ and $F = I\ell B\sin\theta$, and applications such as DC motors, particle accelerators and velocity selectors. Papers usually ask students to explain why a particle curves or a conductor moves, to give a force direction or sketch a path, and sometimes to calculate a force or derive a relationship.

Recognise: A charged particle or wire sitting in a magnetic field, a question asking for the direction of the force or the path a particle takes, a rotating coil or motor loop, a cyclotron, or a velocity selector. Command words such as 'explain', 'describe the effect' and 'sketch the path' signal that most marks are for reasoning and diagrams rather than arithmetic.

Steps

1. Identify the charge (sign and size) or the current and the length of wire inside the field, and note the field strength B.
2. Find the angle θ between B and the velocity v (or between B and the conductor ℓ). The force is largest at 90° and zero when the motion is parallel to the field.
3. Calculate the magnitude: $F = qvB\sin\theta$ for a moving charge, $F = I\ell B\sin\theta$ for a wire. Keep the units consistent (metres, teslas, amperes) and give the answer in newtons.
4. Find the direction with the right-hand rule: the force is perpendicular to both v (or ℓ) and B. Reverse it for a negative charge.
5. For circular motion, equate the magnetic force with the centripetal force, $qvB = mv^2/r$, so $r = mv/(qB)$. Use this for path radius, cyclotron frequency, or the velocity selector, where $qvB = qE$ gives $v = E/B$.
6. For a motor loop, the forces on opposite sides act in opposite directions to form a couple; torque is force multiplied by the perpendicular distance from the pivot.

Where the marks are

- Direction is usually a separate mark and must be stated in words such as up, down, left, into or out of the page; a correct direction can score even if the magnitude is wrong.
- Correct substitution into the force formula including the sinθ term, correct arithmetic and correct units are often awarded as separate marks.
- On path diagrams, each particle's curve is marked separately: it must bend toward the side the force acts on, and straight-line paths do not earn the mark.
- Explanation answers are scored by key points: the force arises because a moving charge is in a field, the force is perpendicular to the velocity, and its effect on the path or speed follows from that.
- Derivations score a mark for equating the magnetic force with another force (usually centripetal or electric) and a further mark for rearranging to the required form.

Pitfalls

- Taking θ as the angle to the wrong direction or dropping the sinθ term, so the result is too large when the motion is not perpendicular to the field.
- Forgetting that a charge moving parallel to the field feels no force, and so does not curve.
- Using the same direction for negative and positive charges, or applying the wrong hand in the right-hand rule.
- Saying the field 'pushes' or 'pulls' the particle without stating that the force is perpendicular to the velocity, so the speed is unchanged; this loses explanation marks.
- Leaving lengths in centimetres when substituting into $F = I\ell B$, omitting units, or drawing a path that bends away from the force or stays straight.

Pattern 11: Torque in a DC motor

Syllabus: S26-U4-SU-11 / S27-U4-SU-11

Frequency: 3 of 10 exam years; 15 marks in all.

Marks reward: Final answer 7, Substitution and working 3, Units, significant figures, direction 2, Qualitative explanation 2, Select the principle or relationship 1

Past questions: 2016R-10；2023R-1(a)；2023R-1(b)；2025R-4；2025R-8

Summary: This point covers the torque on a current-carrying coil in a DC motor: the magnetic force on each current-carrying side produces a turning effect about the axis, and the torque depends on the lever arm and the angle between the force and that lever arm. Papers usually give a calculation with a current, field, length

or turns, and a coil angle, then ask for magnitude and direction, sometimes with a short explanation of rotation or back emf.

Recognise: Look for a DC motor or coil in a magnetic field, a wire at right angles to the field, a pivot or axis, and a lever arm (the perpendicular distance from the axis to the force). Any question asking for torque, turning direction, or how the coil rotates is on this point. Watch for a coil angle given to the horizontal or vertical: convert it to the angle between the force and the lever arm before using sine.

Steps

1. Identify the force on each side of the coil using F = BIl (or F = nBIl for n turns), with the force perpendicular to both the wire and the field.
2. Identify the lever arm r: the perpendicular distance from the pivot to the line of action of the force.
3. Find the angle θ between the force and the lever arm, converting from any angle given to the horizontal or vertical, since a complement or supplement may be needed.
4. Apply the torque relation τ = rF sin θ. For a rectangular coil with two sides producing torque, multiply by two and by the number of turns if given.
5. Substitute with consistent SI units (A, T, m, N) and give the answer in N m, with a sign or direction if asked.
6. Determine rotation direction using the right-hand rule or Fleming's left-hand rule on one side of the coil, and state clockwise or anticlockwise as viewed from the stated direction.

Where the marks are

- Marks are given for the correct force-to-torque chain: a mark for F = IlB, and a mark for substituting into τ = rF sin θ with r and θ correctly identified.
- Marks are given for handling the angle correctly, including converting an angle given to the horizontal or vertical to the angle with the lever arm.
- Marks are given for counting both sides of the coil, and for turns if the question includes them.
- A mark is given for a final value with the correct unit (N m), and for a clear statement of direction when asked.
- Marks are given for a short explanation of rotation, such as the force on a current-carrying side causing the coil to turn in the field.

Pitfalls

- Using the given angle directly in sin θ when the question measures it from the horizontal or vertical, instead of the angle between force and lever arm.
- Forgetting to multiply by two for both sides of the coil, or forgetting the number of turns.
- Mixing up cm and m for lengths, or leaving a value in unconverted units.
- Stating a direction without specifying the viewpoint, or applying the right-hand rule with the wrong orientation of current or field.
- Writing a torque answer without the unit N m, or giving a force answer where torque is required.

Pattern 12: Motional emf of a moving conductor

Syllabus: S26-U4-SU-12 / S27-U4-SU-12

Frequency: 4 of 10 exam years; 15 marks in all.

Marks reward: Final answer 6, Substitution and working 4, Select the principle or relationship 4, Qualitative explanation 1

Past questions: 2016P-16(d); 2022R-5(a); 2022R-5(b); 2023R-12; 2025R-4; 2025R-9

Summary: This point covers the emf induced across a straight conductor that moves through a magnetic field and cuts flux lines, using the relationship emf = ℓvB sinθ. It usually appears as a short calculation, as a force-balance derivation on a sliding bar or wire, or as a qualitative explanation of back emf in a motor.

Recognise: A conductor (a wire, rod, bar, or part of an aircraft or rotating coil) moves at a stated speed through a field, or a bar slides down rails connected through a resistor. Look for wording such as cutting flux

lines, induced emf, potential difference, back emf, constant velocity, or a magnetic force balancing gravity.

Steps

1. Write down the effective length ℓ of conductor in the field, the speed v, and the field strength B, and convert everything to SI units (m, m s-1, T) before substituting.
2. Find the angle θ between the velocity and the field, or resolve B into the component perpendicular to the motion (B sinθ, or B cos if the angle is given to another direction). Only that perpendicular component contributes.
3. Calculate the emf with emf = ℓvB sinθ. The emf is largest when the motion is perpendicular to the field, so sinθ = 1 in that case.
4. If the conductor forms part of a closed circuit of resistance R, find the current with Ohm's law, I = emf/R.
5. For a force-balance question, the magnetic force on the current-carrying conductor is F = BIℓ (using the same perpendicular component). Set it equal to the opposing force, such as mg or mg sinθ, then substitute I = emf/R and emf = ℓvB sinθ and rearrange for the required quantity.
6. For an explanation, state that the moving conductor cuts flux lines so an emf is induced; then state that by Lenz's law (or the right-hand rule) the induced emf or current opposes the motion, which is the back emf in a motor.
7. Give the final answer with correct units (V, A, or m s-1) and a sensible number of significant figures.

Where the marks are

- A mark for converting all quantities to SI units and using the correct field component perpendicular to the motion.
- A mark for writing the emf relationship and substituting values before the final number is reached.
- A mark for linking the emf to current through Ohm's law when the circuit is closed.
- A mark for a correct force balance (magnetic force equals the opposing force), with the magnetic force written in terms of the induced current.
- A mark for an explanation that names cutting flux lines and states that the induced emf opposes the motion (back emf).

Pitfalls

- Using the full field strength when only its component perpendicular to the motion counts, or leaving out sinθ or using cosθ by mistake.
- Substituting km h-1 or cm without converting to m s-1 or m, which gives a wrong answer even with correct method.
- Treating a moving conductor in a closed circuit as carrying zero current, or confusing emf (volts) with current (amperes).
- In force-balance problems, using the wrong length or ignoring the angle of an incline, or setting up the balance with the wrong forces.
- Giving an explanation that only says the conductor moves, without saying it cuts flux lines and that the emf opposes the motion; or omitting units from the final answer.

Pattern 13: Magnetic flux

Syllabus: S26-U4-SU-13 / S27-U4-SU-13

Frequency: 1 of 10 exam years; 7 marks in all.

Marks reward: Graph work 4, Select the principle or relationship 1, Substitution and working 1, Final answer 1

Past questions: 2024P-17(a)； 2024P-17(b)

Summary: This point tests magnetic flux, Φ = B A⊥, for a flat coil or loop in a magnetic field. Papers usually ask for a maximum or specific flux value from given B and area, and then a sketched graph of flux against time as the coil rotates through one full cycle.

Recognise: Look for a coil, loop or frame that rotates or changes orientation in a uniform magnetic field, with

the field strength and coil area given. The question asks for the flux at a stated or maximum orientation, or for a flux-time graph. Watch for the phrase 'perpendicular to the field' or a diagram showing the starting angle.

Steps

1. Identify the field B and the area of the loop, and note the angle between the field and the normal to the loop.
2. Use $\Phi = B A_\perp$, where $A_\perp$ is the component of the area perpendicular to B. When the coil is perpendicular to the field, $A_\perp$ equals the full area A, so $\Phi = BA$ is the maximum flux.
3. For a general orientation with angle θ between B and the normal to the coil, $A_\perp = A \cos\theta$, so $\Phi = BA \cos\theta$. Use this for flux at other angles.
4. Check units: flux is in weber (Wb), and 1 Wb = 1 T m^2. Convert any area given in cm^2 to m^2 before substituting.
5. For a uniformly rotating coil in a uniform field, the flux varies sinusoidally with time. The maximum value is BA and the flux passes through zero when the coil is parallel to the field.
6. Sketch the flux-time graph: a smooth sine (or cosine) curve with amplitude equal to your maximum flux, and one full cycle per rotation. The period on the time axis equals the time for one complete rotation, and the starting point is set by the diagram's starting position.

Where the marks are

- Marks are commonly given for stating the flux equation and correctly identifying the perpendicular area as the one that gives maximum flux.
- A mark is usually given for substituting correct values with units into the equation, which often shows the substitution step separately from the final answer.
- A mark is commonly given for the final numerical answer with the correct unit (Wb) and a sensible number of significant figures.
- Graph marks are usually split between the sinusoidal shape, showing one complete cycle, labelling the maximum flux on the y-axis, and showing the period correctly on the x-axis.
- The starting position of the coil decides where the curve begins (at a maximum or at zero), so a mark may depend on matching the diagram's starting point.

Pitfalls

- Using the full coil area when the coil is not perpendicular to the field, instead of $A_\perp = A \cos\theta$.
- Forgetting to convert area from cm^2 to m^2 before calculating flux, which gives a wrong power-of-ten answer.
- Drawing the flux graph starting at the wrong point, for example starting at zero when the diagram shows the coil perpendicular to the field.
- Plotting the graph with the wrong period, or drawing more or fewer than one complete cycle when the question asks for one rotation.
- Stating the unit as T instead of Wb, or failing to label the axes with the quantity and unit.

Pattern 14: Faraday's law and generators

Syllabus: S26-U4-SU-14 / S27-U4-SU-14

Frequency: 8 of 10 exam years; 66 marks in all.

Marks reward: Qualitative explanation 23, Graph work 16, Final answer 12, Substitution and working 6, Recall 3, Select the principle or relationship 3, Units, significant figures, direction 3

Past questions: 2016P-16(a); 2016P-16(b); 2016R-8; 2017R-8; 2018R-7(a); 2018R-7(b); 2019P-18(a); 2019P-18(b); 2019R-3; 2020C-21(b); 2020P-16(a); 2020P-16(b); 2020P-16(ci); 2020P-16(cii); 2020R-7(a); 2022P-16(c); 2022P-16(d); 2022R-3; 2023R-7(b); 2023R-7(c); 2024P-17(c)

Summary: This point covers electromagnetic induction: a changing magnetic flux through a coil induces an emf, and the same idea underlies DC and AC generators. Papers use it in short explain questions about why

current is produced, in calculations of peak and rms emf for a rotating coil, and in sketching emf-time graphs for a rotating or moving coil.

Recognise: Look for a coil or loop rotating, moving, or sitting in a changing field, a generator or alternator diagram, a magnet moved into or out of a solenoid, or a question asking for the emf, the induced current direction, or a graph of emf against time. Words such as flux, rate of change, rotation frequency, rpm, and rms are strong signals.

Steps

1. Identify what changes: the field B, the area A perpendicular to the field, or the angle between the coil and the field, so that the flux changes.
2. For a general change use emf = -N(ΔB·A⊥)/Δt, or equivalently emf = -N ΔΦ/Δt; the magnitude depends on the rate of change, not the size of the flux itself.
3. For a rotating coil use emf_max = 2πNBAf, with f the rotation frequency in hertz, and convert rpm to Hz by dividing by 60 and cm to m before substituting.
4. Convert peak to rms with emf_rms = emf_max/√2 only when the question asks for rms; do not apply it to a peak value already requested.
5. For a graph, the emf is zero when the coil plane is perpendicular to the field (flux changing slowest) and largest when the coil plane is parallel to the field; the emf-time curve is a sine or cosine shape with period equal to one rotation, and its sign reverses as the flux direction changes.
6. For a direction or Lenz's law question, state that the induced current opposes the change in flux that produces it, then apply it to the approaching or receding pole.

Where the marks are

- Marks are commonly given for naming the principle (changing flux induces emf) before any working.
- A mark is typically awarded for writing the correct equation and stating what each symbol represents, such as the number of turns, field strength, and area perpendicular to the field.
- Unit conversions (rpm to Hz, cm to m, mT to T) are often awarded separately, so show them explicitly.
- Graph questions commonly award marks for the correct shape, correct period, labelled maximum value, and the correct phase relative to the flux curve.
- Explanation marks usually need a causal chain: change in flux, rate of change, induced emf, and then the effect on current or work done.

Pitfalls

- Using the magnitude of flux instead of its rate of change, which gives zero emf when the flux is momentarily constant.
- Forgetting to convert rpm to Hz, so the frequency is 60 times too large.
- Applying the rms conversion to the wrong quantity, or dividing by root two when going from rms to peak.
- Sketching the emf curve in phase with the flux curve instead of a quarter cycle out of phase, or omitting the time axis scale.
- Writing that a larger emf is caused by a larger flux rather than by a faster rate of change of flux, which loses the explanation mark.

Pattern 15: Transformers and power distribution

Syllabus: S26-U4-SU-15 / S27-U4-SU-15

Frequency: 5 of 10 exam years; 32 marks in all.

Marks reward: Select the principle or relationship 9, Final answer 8, Qualitative explanation 7, Substitution and working 5, Recall 3

Past questions: 2018R-1(a); 2018R-1(b); 2020C-21(ai); 2020C-21(aii); 2022C-18(c); 2022P-16(a); 2022P-16(b); 2022P-16(c); 2022P-16(d); 2024R-9(a); 2024R-9(b); 2024R-9(c); 2024R-9(d); 2025C-19(c); 2025C-19(d)

Summary: This point covers step-up and step-down transformers in large-scale AC power transmission and distribution, using the turns-ratio relation, power conservation in an ideal transformer, and power loss in transmission lines (P = VI = I^2R = V^2/R). Papers usually pair short calculations of voltage, turns ratio, current, line loss or efficiency with written explanations of why voltage is stepped up before long-distance transport and where transformers sit in the grid.

Recognise: Look for a transformer on a grid diagram (generation, transmission, distribution, end user), a question giving primary and secondary turns or voltages, a long transmission line with resistance per unit length, a request for efficiency or power lost in the wires, or a prompt asking why high voltage is used for transmission. Questions that mention a laminated iron core and eddy currents also belong here.

Steps

1. Identify which coil is primary and which is secondary, and decide whether the transformer is step-up (secondary voltage larger) or step-down (secondary voltage smaller) from the grid position.
2. Set up the turns relation Vp/Vs = Np/Ns, keeping primary on top, and solve for the unknown voltage or turns ratio.
3. For an ideal transformer, assume power in equals power out: Pp = Ps, so Vp Ip = Vs Is. Rearrange to find the unknown current.
4. For a transmission line, find the total resistance as resistance per metre times the full length of the circuit (both conductors, so double the one-way length if needed). Convert units carefully.
5. Find power lost in the line using P_lost = I^2 R, where I is the current in the line, which is also the value that appears in P = VI.
6. Efficiency = (useful output power / input power) x 100%, or equivalently (1 - P_lost / P_input) x 100%. Check the answer is less than 100% unless the question states an ideal transformer.
7. For explanations, build a causal chain: for the same power, a higher voltage gives a lower current; lower current gives a smaller I^2 R loss, so more energy reaches the consumer.

Where the marks are

- Usually one mark for the correct setup of the equation (ratio, power conservation or P = I^2 R) and a separate mark for the substituted numerical result with units.
- Calculation marks depend on correct sequencing: a line-resistance mark, then a power-loss mark, then an efficiency mark, each able to be earned independently.
- Explanation marks reward a named physical link (higher voltage, lower current, less heat in the wires) rather than a general statement that energy is saved.
- Naming the physical cause of core loss (eddy currents or heating) and what laminations do to it earns the mark, even if the wording is brief.
- Stating that transformers need a changing (AC) current to produce a changing magnetic flux and an induced EMF is typically one mark for the principle.

Pitfalls

- Inverting the turns ratio, for example using Np/Ns when the question needs Vs/Vp, which swaps step-up and step-down answers.
- Using the supply or generator voltage in P = I^2 R instead of the current flowing through the line.
- Forgetting that the line has two conductors, or mixing per-metre and per-kilometre resistance units.
- Writing that energy is lost without saying it is converted to heat in the wires, the core, or as eddy currents.
- Claiming a transformer works with DC, or treating an efficiency result over 100% as acceptable.

Pattern 16: Lenz's law

Syllabus: S26-U4-SU-16 / S27-U4-SU-16

Frequency: 7 of 10 exam years; 17 marks in all.

Marks reward: Qualitative explanation 8, Units, significant figures, direction 3, Recall 3, Graph work 1, Select

the principle or relationship 1, Final answer 1

Past questions: 2016P-16(c); 2017R-8; 2019P-18(b); 2020P-16(ci); 2020R-7(a); 2021R-10(a); 2021R-10(b); 2023R-7(a); 2023R-7(b); 2025R-4

Summary: Lenz's law, derived from conservation of energy, fixes the direction of an induced current: it always acts to oppose the change that produced it. Papers test it mostly as a written explanation (stating the law, then linking the induced current to a force, a braking effect, or energy transfer), with some direction-of-current or graph-of-emf questions.

Recognise: Look for words such as explain, direction of induced current, why a magnet slows in a pipe or disc, why a coil or motor needs more force, back emf, or a galvanometer needle deflecting. Any question that asks why something resists motion, or which way current flows when flux changes, is a Lenz's law question. Questions that ask for the size of the emf usually need Faraday's law as well.

Steps

1. Identify the change in magnetic flux that is causing the induced current (a magnet moving in or out, a coil rotating, a loop leaving a field).
2. State the rule: the induced current flows in the direction that creates a magnetic field opposing that change in flux.
3. Use the induced field to name a pole or a force: for example, approaching magnet means the coil produces a like pole facing it, so the magnet is repelled.
4. Use the right-hand grip rule on the coil or wire to convert the induced pole or field direction into the direction of current (clockwise or anticlockwise, left to right or right to left).
5. Link to conservation of energy: the opposing force does work against the motion, so kinetic or mechanical energy is converted into electrical energy and then heat.
6. State the consequence plainly: a retarding force, a slowing of the disc or magnet, or an extra force needed to keep a motor or generator turning at constant speed.
7. For a motor, state that the back emf opposes the supply emf, reducing the net current in the coil.

Where the marks are

- Stating Lenz's law in full, including the idea of opposing the change in flux, not just the direction of current.
- Naming the effect of the induced current: an opposing force, a braking effect, or an extra force needed.
- Linking the effect to energy: electrical energy is produced at the expense of mechanical energy, so energy is conserved.
- Giving the direction of current or the pole that is induced, with a clear reason based on opposing the change.
- For graph or direction questions, a mark for each correct sign or direction that follows from the opposing rule.

Pitfalls

- Saying the induced current opposes the magnet or the motion, instead of the change in flux that produced it.
- Describing the induced field as opposing the field itself rather than the change in field.
- Giving the direction of current without linking it to a pole or field, so the reason is missing.
- Forgetting to connect the answer to energy conservation when the question asks for an explanation in those terms.
- Confusing back emf with the applied emf, or saying back emf increases the current rather than reducing it.

Pattern 17: Applications: back emf, eddy currents and more

Syllabus: S26-U4-SU-17 / S27-U4-SU-17

Frequency: 2 of 10 exam years; 7 marks in all.

Marks reward: Qualitative explanation 3, Select the principle or relationship 2, Final answer 2

Past questions: 2018R-2(a); 2018R-2(b); 2025R-4

Summary: This point asks how electromagnetic induction explains three real devices: the back emf in an electric motor, energy recovery in regenerative braking, and eddy-current heating in induction hotplates. It usually appears as a short explanation of how back emf arises, or as a calculation or 'what happens if the motor jams' question that links back emf, current and power.

Recognise: Look for a device that rotates, brakes or heats using a changing magnetic field, and questions that use the words back emf, opposes the applied voltage, regenerative braking, eddy currents or induction hotplate. Any question that asks what happens to current when a motor stops turning is almost always testing back emf.

Steps

1. Name the induction effect: a coil moving in a magnetic field, or a changing field linked with a conductor, induces an emf.
2. For a motor, state that the rotating coil induces a back emf that opposes the supply voltage, following Lenz's law.
3. The current through the winding is set by the net voltage: I = (V - E_back) / R.
4. Mechanical output power is linked to back emf by P_out = E_back x I, so I = P_out / E_back when power and back emf are known.
5. If the motor jams, rotation stops, E_back falls to zero, so the current rises towards V / R; state both the change and this reason.
6. Regenerative braking: the motor works as a generator, the induced emf drives current back into the supply, and kinetic energy is converted to electrical energy. Induction hotplates: an alternating current in a coil gives a changing field that induces eddy currents in the ferromagnetic base, and heating comes from the base's resistance, P = I^2 R.

Where the marks are

- Credit for stating that back emf opposes the applied voltage, not just that it exists.
- Calculation parts usually give one mark for the correct power relation and one for correct rearrangement and substitution with units.
- For a jammed motor, one mark for the direction of the current change and one for the reason linked to the missing back emf.
- Explanation of induction devices earns marks for naming the induced emf or eddy currents and the opposition or energy transfer involved, rather than general descriptions such as 'it makes electricity'.

Pitfalls

- Saying back emf stops the current, rather than reducing the net voltage that drives it.
- Using the supply voltage instead of the back emf in the power relation.
- Thinking a stalled motor draws less current because it is not moving; with zero back emf the current increases.
- Describing induction hotplates as heating by direct contact or by magnetic heating, without eddy currents and resistance in the pan.
- Describing regenerative braking only as 'braking' without saying the energy flows back to the supply as electrical energy.

Pattern 18: Electromagnetism in society: motors, generators, transformers

Syllabus: S26-U4-SHE-01 / S27-U4-SHE-01

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point covers how electromagnetic principles are used in real devices: motors, generators and transformers, power generation and transmission, transport (electric and hybrid vehicles, maglev trains),

communication, MRI, and charged-particle devices such as the linac, cyclotron, synchrotron, mass spectrometer and velocity selector. The past-paper data for this point is empty (parts_total 0), so this note is written from the physics alone and has not been checked against examined questions; questions usually ask you to explain the principle behind a named device and then apply one or two equations to it.

Recognise: A question names a device (motor, generator, transformer, MRI, cyclotron, mass spectrometer, maglev) and asks how it works, why it works, or what happens to a quantity when a design changes. Look for words such as motor, generator, induced EMF, turns ratio, transmission losses, velocity selector, circular path in a magnetic field, or 'explain using the principle of'.

Steps

1. Name the physical principle the device relies on: a force on a current or moving charge (motor, velocity selector, cyclotron, mass spectrometer), electromagnetic induction from changing flux (generator, transformer), or a strong field with resonance (MRI).
2. For a motor, write F = BIL sin(theta) for the force on a current-carrying conductor, use the right-hand rule for direction, and explain that a split-ring commutator reverses the current every half turn so the torque keeps one direction.
3. For a generator, use Faraday's law EMF = -N dPhi/dt, state that rotating the coil changes the flux linkage, and give the sinusoidal output EMF = NBA omega sin(omega t) with peak value NBA omega.
4. For a transformer, use Vp/Vs = Np/Ns and, for an ideal transformer, Vp Ip = Vs Is. Explain that only changing (AC) current produces a changing flux in the core.
5. For transmission, use P = I^2 R for the power lost in the line. Increasing the voltage by a step-up transformer lowers the current for the same delivered power, so losses fall.
6. For charged particles in a magnetic field, equate qvB = mv^2/r to get r = mv/(qB). For a cyclotron, f = qB/(2 pi m) is independent of speed. For a velocity selector, qE = qvB gives v = E/B. For a mass spectrometer, use r = mv/(qB) to separate ions by m/q.
7. For transport, communication and MRI, state the key relationship briefly (e.g. Lorentz force levitating or propelling a train, induced EMF in a receiving coil, precession at the Larmor frequency) and link it back to the principle in step 1.

Where the marks are

- A mark for naming the correct principle (force on a current, induction, or force on a moving charge) before any equation.
- A mark for a correct equation with each symbol identified, and for using the relationship in the required direction.
- A mark for the direction argument, such as using the right-hand rule or stating that the commutator reverses the current.
- A mark for explaining why high voltage reduces transmission losses using I^2 R.
- A mark for linking the principle to the named application, not just restating the device's purpose, and for stating assumptions and units.

Pitfalls

- Mixing up motor and generator: a motor turns electrical energy into motion using a force on a current, while a generator turns motion into EMF using induction.
- Saying a transformer works with DC; only a changing current gives a changing flux.
- Forgetting that a velocity selector only passes particles with v = E/B; the particle must be undeflected, not just in the field.
- Using mass instead of m/q for a mass spectrometer, or forgetting that cyclotron frequency does not depend on speed.
- Writing sin where cos is needed in torque or EMF expressions, or explaining transmission losses as a property of the wire alone without referring to the current that flows through it.